

Skill Match Resume Matcher and Skill Recommender

Infosys Springboard Virtual Internship 6.0 | Milestone

4

Batch 4

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Resume Matcher

Free and Open Source ATS to help your resume pass the screening stage.

Visit the website www.resumematcher.fyi

Give Resume Matcher a Star on [GitHub](https://github.com)

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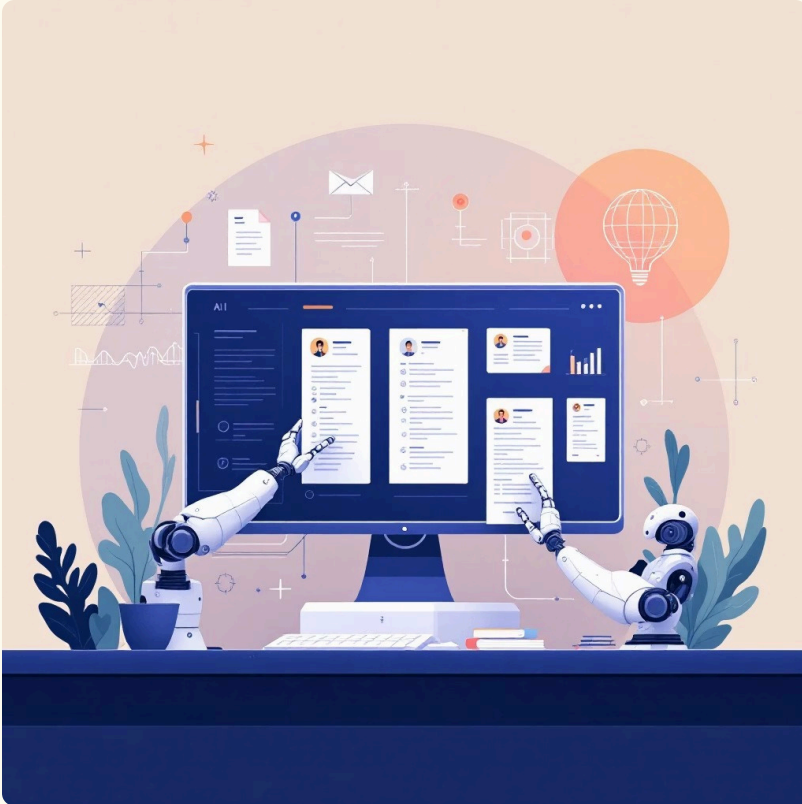
Updates follow me on Twitter.

Follow @srbhr_

If you like the project and would like to further help in development please consider 🍵

Buy me a coffee

Abstract



Automated applicant tracking systems frequently filter out qualified candidates due to inadequate keyword matching and skill alignment. The **Skill Match Resume Matcher and Skill Recommender** addresses this challenge by implementing artificial intelligence and natural language processing to create intelligent connections between job seekers and employment opportunities.

The system performs comprehensive analysis of resumes and job descriptions, providing skill matching, compatibility scoring, and personalised improvement recommendations.



Core Capabilities

Resume parsing, job description analysis, skill extraction and normalisation, intelligent matching engine



Enhanced Features

Visual keyword highlighting, AI-driven suggestions, multi-job optimisation, learning pathway recommendations



Technology Foundation

Python, spaCy, NLTK, scikit-learn, generative AI APIs, Streamlit/React.js, PostgreSQL/MongoDB

System Requirements

1

User Requirements

- Secure authentication (Sign Up/Sign In via Supabase)
- Resume and job description input capabilities
- Real-time analysis and processing
- Actionable feedback highlighting strengths and weaknesses
- Curated course recommendations for skill development

2

System Requirements

- AI integration with Gemini and heuristic fallback
- Strict JSON parsing for AI output validation
- Responsive web interface using Tailwind CSS
- Scalable data storage infrastructure
- Multi-format document processing support



Proposed System Architecture

System Overview

A modern, single-page application (SPA) delivering an intuitive user experience through a streamlined workflow: **Upload/Paste Content** ³ **Intelligent Matching** ³ **Actionable Insights** ³ **Learn & Improve**.



CoreAnalysis Engine

The system features dual analysis modes to ensure reliability and depth:

AI-Powered Analysis

Utilises Google Gemini for deep semantic understanding and contextual skill matching

Heuristic Fallback

Rules-based analysis ensures consistent performance when API access is unavailable

Skill Extraction and Normalization (NLP)

It ensures the extracted skills are normalized which is necessary for the subsequent Intelligent Matching to work effectively.

Key Implemented Features

AI-Powered Analysis

Leverages Gemini AI for contextual skill matching and nuanced evaluation of resume content against job requirements

Match Score & Breakdown

Clear percentagescoringwithvisual charts showing overall compatibility and detailed breakdown

Improvement Checklist

Actionable,specific recommendations for resume content enhancement tailored to target roles

Curated Course Recommendations

Targetedcoursesuggestionsorganised byidentified skill gaps, mapped from `src/lib/courses.ts` to accelerate professional development.

Modern Tech Stack

Built with **React18**, **TypeScript**, **Vite**, and **Tailwind CSS** for a fast, responsive, and maintainable interface.

System Architecture & Technology Stack

Frontend Layer

- React18 with TypeScript
- Vite build tooling
- Tailwind CSS for styling
- Dashboard UI components

Backend & Database

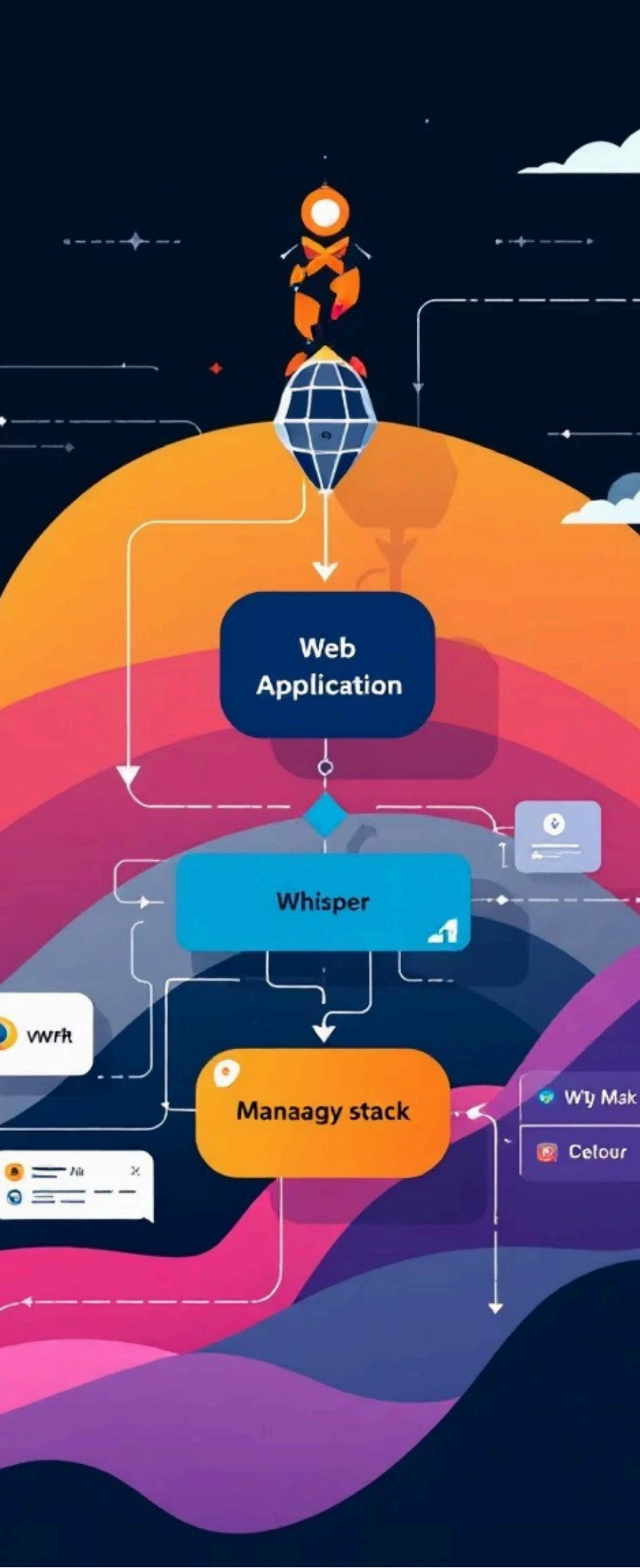
- Supabase JSSDK
- Secure authentication
- Database management
- Real-time data sync

AI/ML Layer

- GoogleGemini API
- NLP libraries (spaCy, NLTK)
- Skill extraction logic
- Semantic analysis

Deployment

- Static host deployment
- Optimised dist/ output
- CDN distribution
- Scalable infrastructure





SkillMatch Workflow



Authenticate (Supabase)

User signs in or signs up using Supabase Auth, gaining secure access to the Dashboard



Input (Dashboard)

User uploads or pastes their resume content and the specific job description text



Analyse (AI/Heuristic)

System uses Gemini AI or heuristic fallback for deep semantic analysis, skill extraction, and normalisation to compare resume against job requirements



Action & Improve (Insights)

Dashboard displays precise match score, visual breakdown of strengths/weaknesses, actionable recommendations, and curated learning paths to close skill gaps

Future Enhancements

Learning Platform Integration

Seamlessly connect with online courses and certification providers to close identified skill gaps efficiently

AI-Powered Interview Preparation

Provide virtual interview simulations with real-time feedback and personalised coaching

Predictive Career Pathing

Utilise advanced analytics to suggest future career trajectories based on market trends and individual strengths

Global Skill Database Expansion

Broaden coverage to include emerging technologies and international job market requirements



Conclusion & Impact

100%

Project Completion

Successfully developed and deployed a fully functional, end-to-end AI-assisted platform

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Core Modules

Resumeparsing, job analysis, skill matching, and learning recommendations

360°
Comprehensive View

Complete analysis from initial upload to actionable improvement strategies

Real-World Impact

SkillMatch **empowers job seekers** to optimise their resumes and learn strategically, focusing time and effort on skills that truly matter for their target roles. By bridging the gap between candidates and opportunities, we're helping individuals **land their dream jobs** whilst streamlining recruitment processes for employers.

Thank You

We extend our sincere gratitude to the **Infosys Springboard team** and our dedicated mentors for their invaluable guidance and support throughout this transformative learning journey.

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